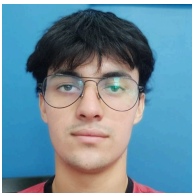


MARTIN POROLLAN

Systems Engineering Student | Full Stack Developer | Data Engineering & AI Automation

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PROFESSIONAL SUMMARY

Fourth-year Systems Engineering student at UTN with C1 English proficiency and over 1.5 years of experience in business process transformation. Specialized in building high-impact automation tools and custom software ecosystems spanning data engineering, applied AI (Vision LLMs, MCP), and industrial IoT. Proven ability to identify operational bottlenecks and solve them using diverse tech stacks, including Java, Python, .NET, and Flutter/Firebase.

PROFESSIONAL EXPERIENCE

Operations & IT Automation Analyst, *[Almacén de agua]*

11/2024 – Present
Mendoza, Argentina

- **Financial Process Engineering (Java):** Architected and deployed a high-reliability Java system for automated payment imputation. The tool reconciles bank statements and manages accounts for a database of 600+ clients, reducing manual processing time by over 50%.
- **Fiscal Automation (Python):** Developed an automation engine in Python to interface directly with the ARCA (ex-AFIP) API. Streamlined the generation of 150+ monthly electronic invoices, eliminating manual entry errors and ensuring 100% fiscal compliance.
- **Invoice Digitization with Vision-Language LLMs (Python):** Built a Python pipeline for ingesting and digitizing purchase invoices (VAT/Compras) using Vision LLMs. The pipeline extracts structured data and automatically generates the files required by the ARCA API/web platform for direct tax imputation.
- **Bank Statement Reconciliation by Tax ID (Java):** Built a Java system for programmatic parsing of bank statements and intelligent reconciliation of client accounts mapped by CUIT (Argentine Tax ID), reducing reconciliation from ~5 hours of manual admin work per week to a simple CSV download.
- **Industrial IoT & Monitoring:** Engineered and deployed an industrial telemetry/IoT system with 2 ESP32 microcontrollers and 4 optical barrier sensors for real-time production counting and continuous capture of operational metrics.
- **Full Stack ERP Development (Flutter):** Designed and launched a comprehensive mobile/desktop ERP using Flutter and Firebase, currently managing 3,600+ digital delivery notes (remitos) over 6 months and tracking real-time stock across multiple product lines.
- **Operations & Logistics Management:** Coordinating 2 daily distribution routes serving 60–100 clients/day and technical service schedules, ensuring SLA fulfillment for a high-volume water distribution facility.
- **Business Intelligence:** Analyzing customer subscription (“abonos”) performance and demand patterns (~3,000 20L bottles/week baseline) to detect churn signals and forecast recurring revenue.

EDUCATION

Bachelor of Science in Systems Engineering (4th Year Ongoing), *National Technological University (UTN)*

Present
Mendoza, Argentina

English Language Studies, *Instituto Amicana*

- TOEFL iBT — C1 Level Achieved

2015 – 2019
Mendoza, Argentina

TECHNICAL SKILLS

Languages

- Java
- Python

Data, AI & Automation

- Vision-Language LLMs (VL-LLMs)
- Model Context Protocol (MCP)

- Dart (Flutter)
- JavaScript / TypeScript (Node.js, React)
- C# (.NET 8)

Back-end & Databases

- Firebase (Firestore, Auth)
- PostgreSQL
- RESTful APIs
- API Testing (Postman)
- SQL / NoSQL

DevOps & Tools

- Linux (Ubuntu, Zorin OS)
- Git
- Docker
- Kubernetes
- CI/CD (GitHub Actions)
- Cloud (AWS, Azure)

- Local LLM implementation (Ollama, Qwen, Gemma)
- Process & Fiscal Automation

IoT & Industrial Automation

- ESP32 Microcontrollers
- Industrial Telemetry & Sensors
- PLC Programming
- Real-time Monitoring Systems

Operations

- ERP/CRM Architecture
- Fiscal API Integration (ARCA/AFIP)
- Logistic Optimization

LANGUAGES

English — C1 Advanced/Fluent

Spanish — Native

KEY PROJECTS

Water Logistics & Automation Ecosystem, Lead Developer

- **Payment Reconciliation Engine:** Java-based financial tool for mass account reconciliation by CUIT.
- **ARCA Invoicing Bot:** Python script for automated government-level billing.
- **Invoice Digitization Pipeline:** Python + Vision LLM pipeline for automated purchase-invoice ingestion and ARCA imputation.
- **Enterprise ERP App:** Flutter/Firebase cross-platform application for end-to-end business management.
- **Industrial IoT Telemetry:** ESP32-based sensor network for real-time production counting and metrics.

ADA Application Suite, Lead Architect & Developer

- **Software Ecosystem Architecture:** Architected and developed a comprehensive business-management software ecosystem using .NET 8, PostgreSQL, React, and Flutter, comprising a Self-Service Client Portal, a Factory & IoT App, and a Field Service & QR Traceability App.
- **Self-Service Client Portal:** Platform for customers to self-manage orders, account status, and billing.
- **Factory & IoT App:** Application connected directly to the plant's sensor network for real-time production monitoring and control.
- **Field Service & QR Traceability App:** Linked to the self-service portal for field maintenance management, with each dispenser uniquely identified via QR code.
- **AI Integration with MCP:** Interconnected applications and databases via the Model Context Protocol (MCP), enabling semantic analysis and dynamic, personalized reporting for decision-making.
- **PLC Programming:** Programmed PLCs (Programmable Logic Controllers) for industrial process control and automation.

TECHNICAL INTERESTS & ACHIEVEMENTS

- **AI Implementation:** Experience configuring and running local Large Language Models (Ollama, Qwen, Gemma) and Vision-Language LLMs for internal data automation.
- **Advanced Infrastructure:** Expert-level management of Linux-based development environments.
- **Industrial Automation:** Hands-on experience with IoT telemetry (ESP32) and PLC programming for production monitoring and control.
- **Digital Transformation:** Successfully transitioned a traditional warehouse into a 100% digital, paperless operation.